



Meenakshi Sundararajan Engineering College

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NEARSY

AI-Powered Nearby Service Finder Platform for Quick and Reliable Local Services

VAC PROJECT PRESENTATION

PRESENTED BY:

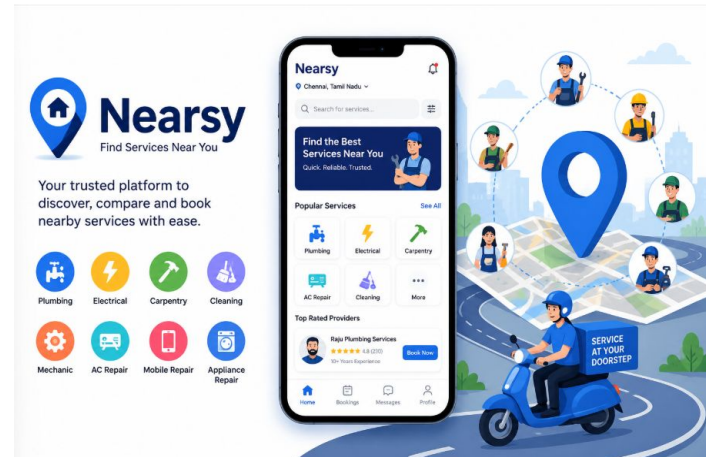
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Introduction

Nearsy is an intelligent nearby service discovery and booking platform that utilizes **Artificial Intelligence** and **Location-Based Services** to connect users with trusted local professionals in real time. The system evaluates **user location, service category, provider availability, customer ratings, and response time** to recommend suitable service providers. Users can quickly search, compare, and book nearby professionals, ensuring convenience, reliability, and a seamless service experience.

Key Features

- ☐ Real-time nearby service discovery
- ☐ AI-based smart recommendations
- ☐ GPS-enabled location tracking
- ☐ Verified service provider profiles
- ☐ Instant online service booking



Finding **trusted local service providers** during emergencies or daily household needs is often time-consuming, making it difficult for users to quickly access **reliable services**. Traditional search methods and online directories often provide outdated information, limited availability, and lack personalized recommendations.

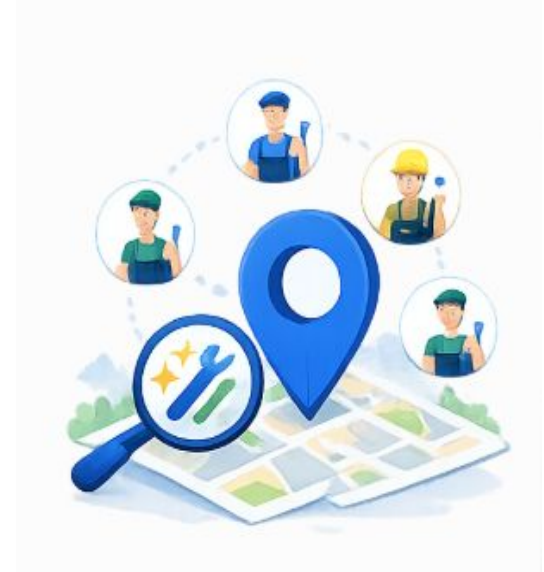
This results in:

- ☐ Difficulty finding trusted professionals
- ☐ Delayed service availability
- ☐ Poor service quality and reliability
- ☐ Limited information about providers
- ☐ Increased time and effort
- ☐ Inconvenient booking experience



An **AI-driven** nearby service discovery platform is required to accurately recommend trusted service providers based on **location, availability, and customer ratings**, while enabling fast, secure, and hassle-free service booking.

- ❖ Collect **user location** and nearby service provider data.
- ❖ Process **service requests** for intelligent recommendations.
- ❖ Analyze **user preferences** and provider availability in real time.
- ❖ Detect suitable service providers using **AI models**.
- ❖ Generate **personalized service recommendations**.
- ❖ Verify **service provider details** before booking.
- ❖ Reduce **service search time**.
- ❖ Improve **recommendation accuracy**.
- ❖ Enhance customer convenience and **service reliability**.
- ❖ Continuously learn from user feedback and **booking patterns**.



S. No.	Paper / Authors	Year	Methodology / Approach	Advantages	Disadvantages
1.	AI-Based Recommendation System for Local Service Discovery	2024	Uses Artificial Intelligence to recommend nearby service providers based on user location, preferences, ratings, and service history.	- Personalized recommendations - Improved user experience - Faster service discovery	- Requires large user datasets - Recommendation accuracy depends on data quality
2.	Location-Based Service Recommendation Using Machine Learning	2024	Applies Machine Learning with GPS and user behavior analysis to identify the most suitable nearby services.	- Accurate location-based results - Reduces search time - Better recommendation quality	- GPS accuracy limitations - Performance affected by network availability
3.	Smart Local Service Booking Platform Using Artificial Intelligence	2024	Integrates AI algorithms with online booking to recommend verified professionals based on availability and customer reviews.	- Reliable service providers - Efficient booking process - Improved customer satisfaction	- Requires continuous data updates - Scalability challenges with large user base
4.	AI-Driven Service Provider Recommendation and Booking System	2024	Combines AI, customer feedback analysis, and location intelligence to improve service matching and booking efficiency.	- Intelligent service matching - Enhanced provider visibility - Supports real-time recommendations	- Higher computational requirements - Requires regular model training

Module	Description
User Registration Module	Allows users and service providers to create accounts, log in securely, and manage their personal profiles and service information.
Location & Search Module	Detects the user's current location, searches nearby service providers, and filters results based on category, distance, and ratings.
AI Recommendation Module	Uses Artificial Intelligence to analyze user preferences, service history, ratings, and availability to recommend the best service providers.
Booking & Scheduling Module	Enables users to book services, schedule appointments, track booking status, and receive booking confirmations instantly.
Review & Feedback Module	Allows customers to rate completed services, submit feedback, and helps improve future recommendations and provider reliability.

Concepts

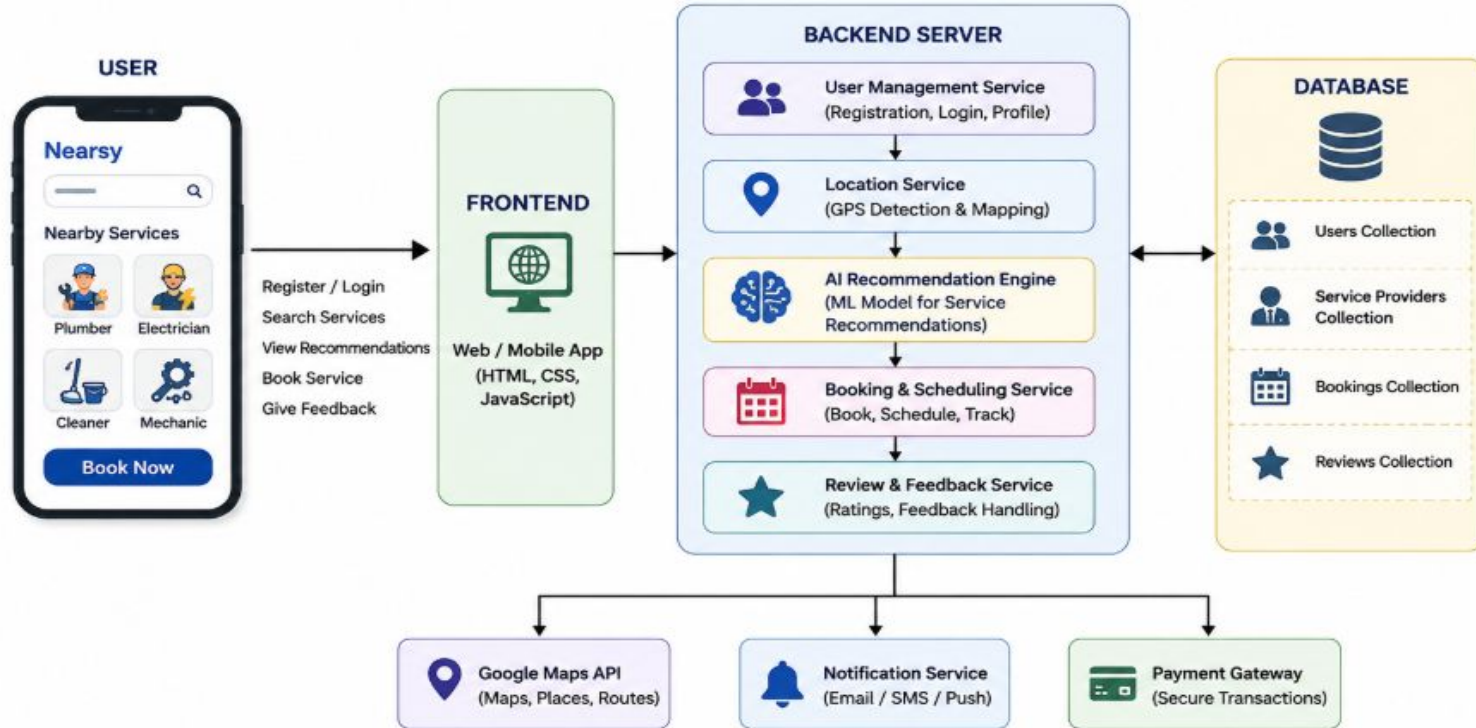
- Artificial Intelligence
- Machine Learning
- Location-Based Services (GPS)
- Service Recommendation
- Smart Booking
- Customer Feedback

Methodology

- Collect user and service provider data.
- Detect user location using GPS.
- Process service requests.
- Generate AI-based recommendations.
- Book nearby service providers.
- Collect ratings and feedback.

Tools used

- HTML, CSS & JavaScript
- Node.js and Express
- MongoDB
- Google Maps API
- Visual Studio Code (VS Code)



Optimization

Performance Optimization

- Optimized location search reduces response time.
- AI-based recommendations improve service matching speed.



Database Optimization

- Centralized MongoDB storage eliminates duplicate records.
- Database validation ensures consistent and reliable data.
- Efficient query execution improves overall performance.



Workflow Optimization

- Automated booking reduces manual service requests.
- Admin dashboard streamlines user and provider management.
- Digital service booking minimizes paperwork and scheduling delays.



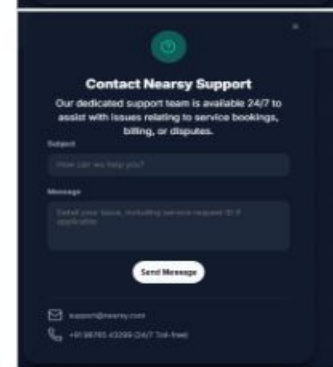
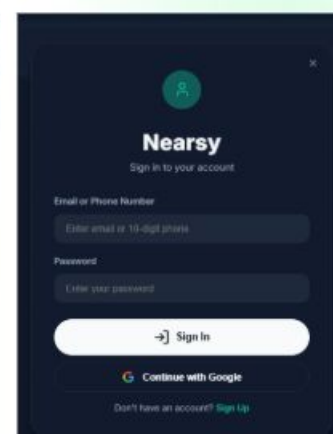
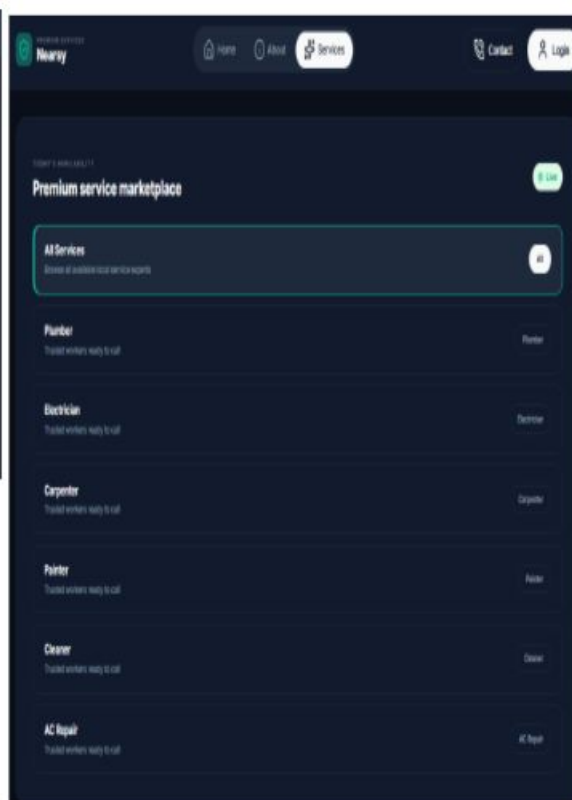
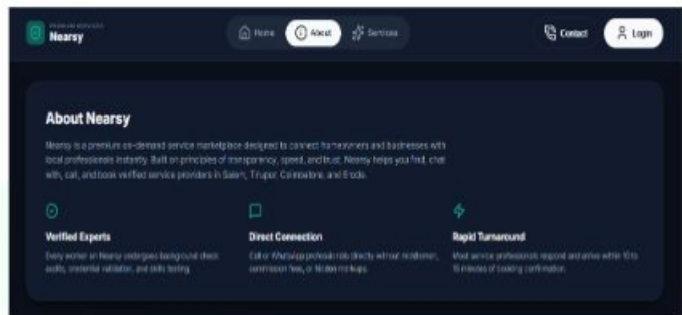
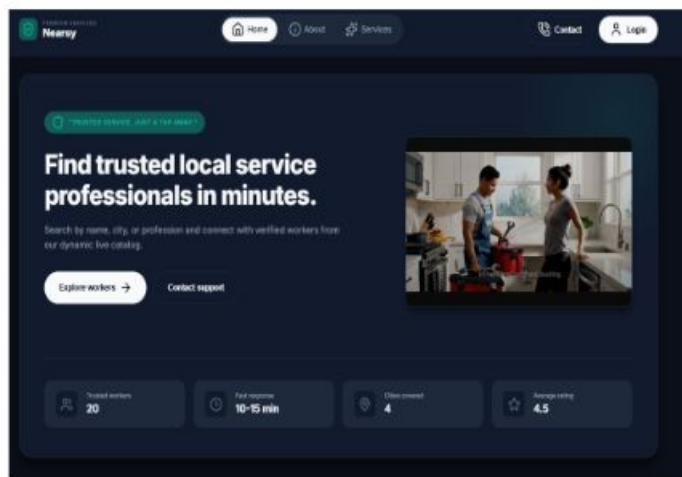
Scalability & Security

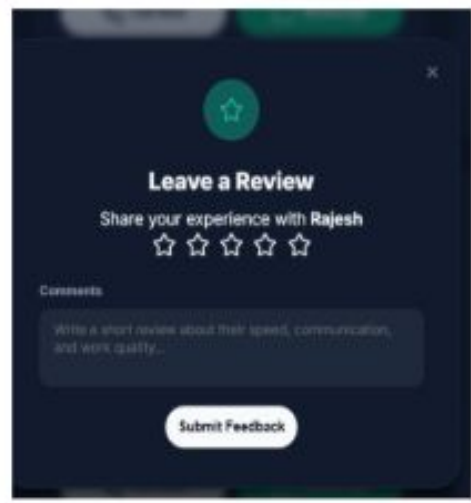
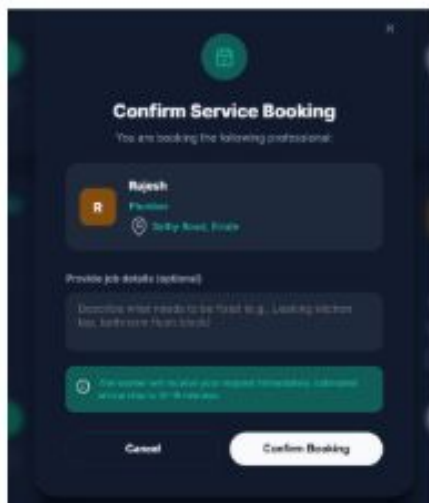
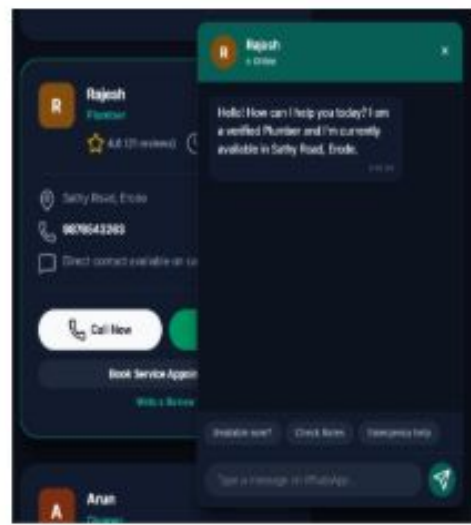
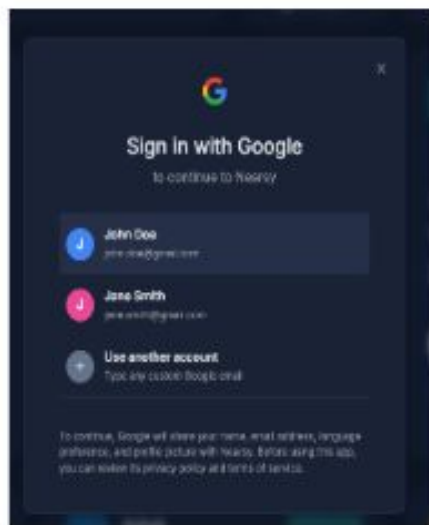
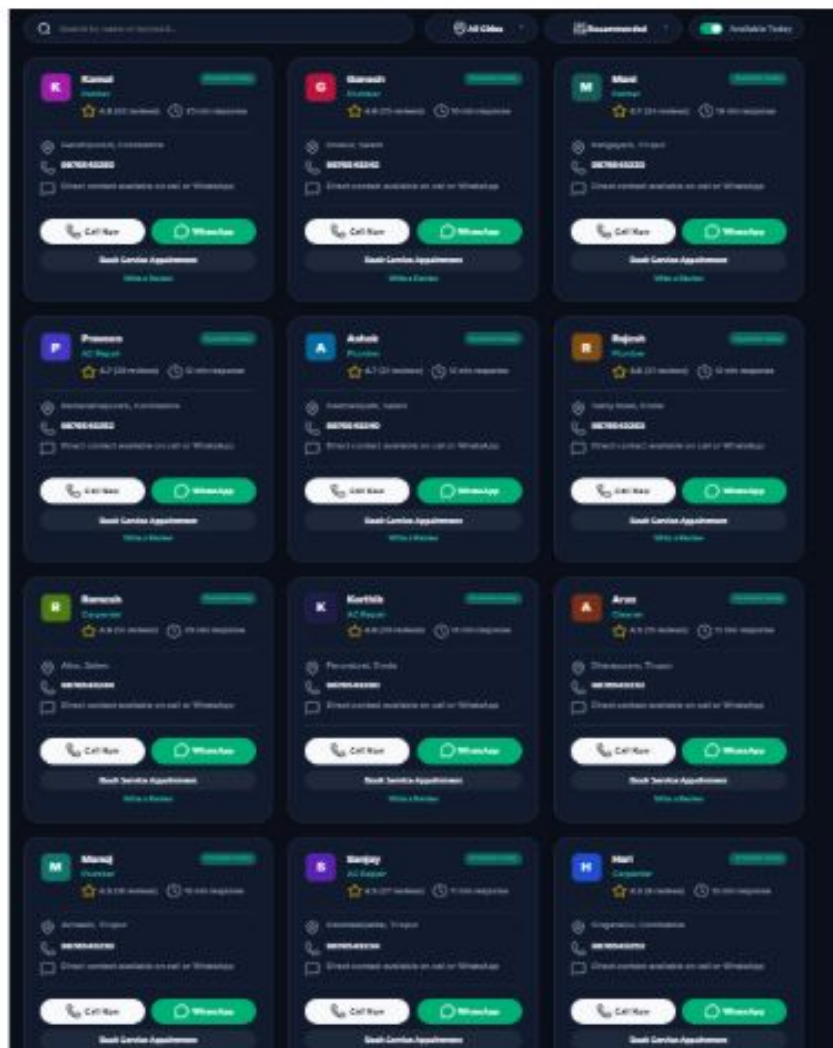
- Modular client-server architecture supports future feature expansion.
- Secure API communication protects user information.
- Cloud database enables reliable access and easy scalability.



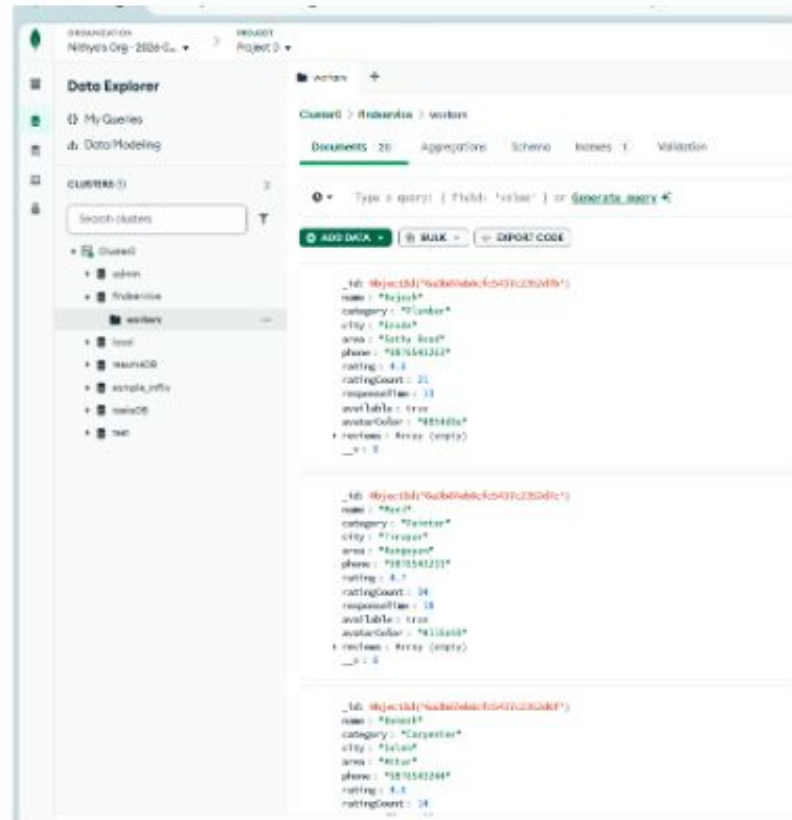
Results and Conclusion

08





```
PS C:\Users\Krish\OneDrive\Desktop\Find service> cd Backend
PS C:\Users\Krish\OneDrive\Desktop\Find service\Backend> node server.js
injected env (2) from .env // tip: % override existing { override: true }
Server running on http://localhost:5000
MongoDB Atlas Connected
```



✔ Successfully retrieved 20 worker documents from MongoDB Atlas cloud.

CONCLUSION

Nearsy provides an intelligent and user-friendly platform for discovering and booking nearby service providers using **Artificial Intelligence** and **Location-Based Services**. By analyzing user location, service availability, customer ratings, and preferences in real time, the system delivers accurate service recommendations and simplifies the booking process. The integration of AI-based recommendations enhances customer convenience, reduces service search time, and improves provider visibility. **Nearsy** offers a reliable, scalable, and efficient solution for connecting users with trusted local professionals, ensuring a seamless and satisfactory service experience.

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- [3] S. Gupta, A. Sharma, and P. Verma, "**Artificial Intelligence for Smart Service Recommendation Systems**," *International Conference on Artificial Intelligence and Data Engineering (AIDE)*, IEEE, 2024.
- [4] D. Singh, K. Rao, and V. Patel, "**Personalized Recommendation Systems: A Review**," *IEEE Access*, vol. 11, pp. 96541–96567, 2023.
- [5] M. Alotaibi, R. Ahmed, and S. Khan, "**Location Intelligence and Recommendation Systems in Smart Cities**," *IEEE Internet of Things Journal*, vol. 11, no. 3, pp. 4521–4535, 2024.
- [6] T. Brown and P. Wilson, "**AI-Driven Local Service Discovery and Booking Platform**," *arXiv preprint*, 2024 (arXiv:2405.10234).